Exercises
Probabilistic Artificial Intelligence
Fall 2025

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For questions, please refer to Moodle.
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Homework 1
(Probability, Bayesian Linear Regression)

General Instructions

- Submission of solutions is not mandatory but solving the exercises are highly recommended. The master solution will be released after the exercise deadline.
- Part of the exercises are available on Moodle as a quiz. These problems are marked with [☑].

Exercise 0: Warm-Up

Throughout this homework p denotes a probability mass function or density, and $\mathcal{N}(\mu, \Sigma)$ a Gaussian with mean μ and covariance Σ.

(a) Let W, Z be continuous random variables and suppose the conditional density of W does not depend on the value of Z, i.e. $p(W \mid Z = z) = q(W)$ for every $z \in \mathbb{R}$ with $p_Z(z) > 0$. Show that $p(W) = q(W)$.

(b) Let f be a density on $\mathbb{R}$ with $f(x) \propto \exp\left(-\frac{1}{2}ax^2 + bx\right)$ for $a > 0$ and $b \in \mathbb{R}$. Which distribution is f?

(c) Now let f be a density on $\mathbb{R}^d$ with $f(x) \propto \exp\left(-\frac{1}{2}x^\top \Lambda x + b^\top x\right)$, where $b \in \mathbb{R}^d$ and Λ lies in the set of symmetric positive definite matrices,

$$\mathbf{S}^d_{++} = \left\{ M \in \mathbb{R}^{d \times d} : M = M^\top \text{ and } v^\top M v > 0 \text{ for all } v \in \mathbb{R}^d \setminus \{0\} \right\}.$$

Which distribution is f? Deduce the value of $\int_{\mathbb{R}^d} \exp\left(-\frac{1}{2}(x-m)^\top \Lambda (x-m)\right) dx$ for an arbitrary $m \in \mathbb{R}^d$.

Exercise 1: True/False Question

(a) Let $X \sim \mathcal{N}(\mu, \Sigma)$ be a Gaussian random vector.

[☑] The marginal distribution of any subset of components of X is not Gaussian. ☐ True ☐ False

[☑] Are linear combinations of jointly Gaussian random variables Gaussian? ☐ True ☐ False

(b) Let X, Y, Z be independent standard normal random variables.

[☑] The random variable $T = \frac{X + e^Z Y}{\sqrt{1 + e^{2Z}}}$ is not Gaussian. ☐ True ☐ False

[☑] Let $\alpha \in \{0, \frac{\pi}{2}\}$ with $\mathbb{P}(\alpha = 0) = \mathbb{P}(\alpha = \frac{\pi}{2}) = \frac{1}{2}$ be independent of X and Y. Then $X\cos(\alpha) + Y\sin(\alpha) \sim \mathcal{N}(0, 1)$. ☐ True ☐ False

[☑] Let $(X, Y) \sim \mathcal{N}(0, \Sigma)$, where $\Sigma = \begin{pmatrix} 1 & \alpha \\ \alpha & 1 \end{pmatrix}$, and $\alpha \in \mathbb{R}, |\alpha| \leq 1$. Then random vectors $U = X + Y$ and $V = X - Y$ are independent if and only if $\alpha = 0$. ☐ True ☐ False

(c) [☑] Let $X_1, \ldots, X_n$ be mutually independent binary random variables. Completely specifying their joint distribution requires $\Theta(2^n)$ parameters. ☐ True ☐ False

[☑] Let $Z \sim \mathcal{N}(\mu, \Sigma)$ be an n-dimensional multivariate Gaussian random variable. We can always find an invertible matrix A such that to completely specify the distribution of AZ, we need $\mathcal{O}(n)$ parameters in the worst case. ☐ True ☐ False

Exercise 2: Multivariate Gaussian Distribution

A vector-valued random variable $x \in \mathbb{R}^d$ is said to have a multivariate normal distribution with mean $\mu \in \mathbb{R}^d$ and covariance matrix $\Sigma \in \mathbf{S}^d_{++}$ if its pdf is:

$$p(x; \mu, \Sigma) = \frac{1}{(2\pi)^{d/2} \det(\Sigma)^{1/2}} \exp\left(-\frac{1}{2}(x-\mu)^T \Sigma^{-1}(x-\mu)\right)$$

Prove the following facts, stated in the first lecture:

(a) Every marginal of a Gaussian vector is Gaussian (slide 51).

(b) Conditioning on a subset of variables of a joint Gaussian is Gaussian (slide 52).

Recall the following fact about characteristic functions, which will help you to solve the next questions: For a random vector $X \in \mathbb{R}^d$, define its characteristic function φ_X as

$$\varphi_X(t) = \mathbb{E}[\exp(i\, t^\top X)], \quad \text{for all } t \in \mathbb{R}^d.$$

The characteristic function completely identifies a distribution. For a multivariate Normal distribution $\mathcal{N}(\mu, \Sigma)$, the characteristic function can be computed explicitly:

$$\varphi(t) = \exp(i\, t^\top \mu - \frac{1}{2} t^\top \Sigma t).$$

(c) [☑] Let $X = (X_1, \ldots, X_d)$ be a d-dimensional standard Gaussian random vector, that is, $X \sim \mathcal{N}_d(0, I)$. Define $Y = AX + \mu$, where A is a $d \times d$ matrix and $\mu \in \mathbb{R}^d$. What is the distribution of Y?

(d) [☑] If B is an $r \times d$ matrix, what is the distribution of BY?

(e) [☑] Let X, Y be i.i.d. $\mathcal{N}(0, 1)$ random variables, then $X + Y$ and $X - Y$ are independent. ☐ True ☐ False

(f) [☑] Let $X = (X_1, X_2)$ be a bivariate Normal random variable with mean $\mu = (1, 1)$ and covariance matrix $\Sigma = \left(\begin{smallmatrix} 5 & 1 \\ 1 & 4 \end{smallmatrix}\right)$. What is the mean and the variance of the conditional distribution of $Y = X_1 + X_2$ given $Z = X_1 - X_2 = 0$?

(g) We consider a dataset $\{(x_i, y_i)\}_{i=1}^n$ of size n, where $x_i \in \mathbb{R}^d$ denotes the feature vector and $y_i \in \mathbb{R}$ denotes the label of the i-th data point. Let ϵ_i be i.i.d. samples from the Gaussian distribution $\mathcal{N}(0, \lambda^{-1})$ for a given $\lambda > 0$. We collect the labels in a vector $\mathbf{y} \in \mathbb{R}^n$, the features in a matrix $X \in \mathbb{R}^{n \times d}$, and the noise in a vector $\epsilon \in \mathbb{R}^n$. The labels are generated according to

$$y = Xw + \epsilon.$$

To perform Bayesian Linear Regression, we consider the prior distribution over the parameter vector w to be $\mathcal{N}(\mu, \lambda^{-1} I_d)$, where I_d denotes the d-dimensional identity matrix and $\mu \in \mathbb{R}^d$ is a hyperparameter.

[☑] The following statement holds for the conditional covariance matrix $\Sigma_y = \text{Var}(y \mid X, \mu, \lambda) = \lambda^{-1} X X^\top + I_n$ ☐ True ☐ False

[☑] Maximum likelihood estimate $\hat{\mu}_1$ of the hyperparameter μ is equal to $\mu_1 = \left(X^\top \Sigma_y^{-1} X\right)^{-1} X^\top \Sigma_y^{-1} y$. ☐ True ☐ False

Exercise 3: Bayesian Inference with Conjugate Distributions

A company wants us to model the average lifetime of their newest LED lightbulbs. For this they observed the lifetime of n bulbs, giving us the dataset $\{x_1, \ldots, x_n\}$ with $x_i \in \mathbb{R}^+$. Due to LED bulbs having a constant failure rate, we assume $x_i \overset{\text{i.i.d}}{\sim} \text{Exp}(\lambda)$. Furthermore, we use a gamma distribution as the prior for the rate parameter λ, i.e., $\lambda \sim \text{Gamma}(\alpha, \beta), \alpha, \beta > 0$. The pdfs of $S \sim \text{Exp}(\lambda)$ and $T \sim \text{Gamma}(\alpha, \beta)$ are, respectively,

$$p_S(s) = \lambda e^{-\lambda s}, \qquad s \geq 0$$
$$p_T(t) = \frac{t^{\alpha-1} e^{-\beta t} \beta^\alpha}{\Gamma(\alpha)}, \qquad t \geq 0,$$

where $\Gamma(\cdot)$ is the gamma function. Let $\hat{\lambda}_{\text{MAP},n}$ be the MAP estimate of λ using the model defined above. Let $\hat{\lambda}_{\text{MLE},n}$ be the MLE estimate.

(a) [☑] What is the likelihood $p\left(x_{1:n} \mid \lambda\right)$?

(b) [☑] What is the posterior distribution of the rate parameter $p\left(\lambda \mid x_{1:n}\right)$?

(c) [☑] What is the MAP estimate of λ, i.e., $\hat{\lambda}_{\text{MAP},n}$?

(d) [☑] If n is sufficiently large, $\hat{\lambda}_{\text{MAP},n} \approx \hat{\lambda}_{\text{MLE},n}$. ☐ True ☐ False

Exercise 4: Online Bayesian Linear Regression

Consider Bayesian linear regression (BLR) with the model

$$w \sim \mathcal{N}(0, I_d), \qquad y_i = x_i^\top w + \varepsilon_i, \qquad \varepsilon_i \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0, \sigma_n^2),$$

where $x_i \in \mathbb{R}^d$, $y_i \in \mathbb{R}$, and the noise variance $\sigma_n^2 > 0$ is known. We condition on the feature vectors, treating them as fixed inputs, and assume that the observation noises are independent of w. Here, I_d denotes the $d \times d$ identity matrix. As seen in the lecture, this model has a Gaussian posterior. If X is the data matrix whose ith row is $x_i^\top$ and y is the vector of responses, then

$$p(w \mid X, y) = \mathcal{N}\Big(w; (X^\top X + \sigma_n^2 I_d)^{-1} X^\top y, (\sigma_n^{-2} X^\top X + I_d)^{-1}\Big).$$

Instead of having all observations available at once, suppose that one new observation (x_t, y_t) arrives at each round $t = 1, 2, \ldots$. Let

$$\mathcal{D}_t := \{(x_i, y_i)\}_{i=1}^t, \qquad p_t(w) := p(w \mid \mathcal{D}_t),$$

with $\mathcal{D}_0 = \varnothing$ and $p_0(w) = \mathcal{N}(w; 0, I_d)$. Our goal is to update the posterior after each observation and to compute the predictive distribution of a new noisy response $y^\star$ at a given feature vector $x^\star$. One approach is to construct the full data matrix $X^{(t)} \in \mathbb{R}^{t \times d}$ and response vector $y^{(t)} \in \mathbb{R}^t$ at every round, then recompute the posterior using the formula above. This gives the correct result, but storing and repeatedly processing all past observations becomes increasingly costly as t grows. Although the matrix to be inverted remains $d \times d$, recomputing its entries from the full dataset requires an increasing amount of work. The following questions first remove the need to retain past observations and then remove the need to recompute a matrix inverse at every round.

(a) Let $X = X^{(t)}$ be the data matrix whose ith row is $x_i^\top$, and let $y = y^{(t)} = (y_1, \ldots, y_t)^\top$. Prove that

$$X^\top X = \sum_{i=1}^t x_i x_i^\top.$$

Derive an analogous expression for $X^\top y$.

(b) Design an algorithm that updates the posterior as observations arrive, using memory independent of t for fixed d. Give its initialization and update rules, explain how to obtain the posterior parameters, and justify the memory requirement. You may still compute a matrix inverse at each round.

Hint: Use part (a) to identify sums that can be updated one observation at a time.

(c) The Sherman–Morrison identity states that, for a symmetric invertible matrix $A \in \mathbb{R}^{d \times d}$ and a vector $x \in \mathbb{R}^d$ such that $1 + x^\top A^{-1} x \neq 0$,

$$(A + xx^\top)^{-1} = A^{-1} - \frac{(A^{-1}x)(A^{-1}x)^\top}{1 + x^\top A^{-1} x}.$$

Prove this identity by multiplying the proposed inverse by $A + xx^\top$.

(d) Computing a matrix inverse at each round costs $O(d^3)$. Modify the algorithm from part (b) to update the posterior parameters in $O(d^2)$ time per round. Specify the quantities to store, their initialization and update rules, and justify the time and memory requirements.

Hint: Apply the identity from part (c) to update the inverse using the new observation.

(e) Using the posterior after observing (x_{t+1}, y_{t+1}) in part (d), derive the predictive distribution of a new noisy response $y^\star$ at a given feature vector $x^\star \in \mathbb{R}^d$.

Express its mean and variance in terms of $x^\star$, $\bar{\mu}_{t+1}$, $\bar{\Sigma}_{t+1}$, and σ_n^2.